Constructor

→ In C++, a constructor is a special member function of a class that gets automatically called when an object of the class is created. It is used to initialized the object's data members.

→ Key features of a constructor :

1.) Has the same name as the class.
2.) It does not have a return type (not even void).
3.) Automatically called when an object is created.
4.) Can be overloaded (multiple constructors with different parameters).

Syntax :-

```
Class class_name
{
    public :        // always public (not private)

    class_name();   // constructor declaration

};
```

{ It is same as function in c }

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→ Types of constructors :-

1.) Default Constructor (No arguments)

2.) Parameterized Constructor (Takes arguments)

3.) Copy constructor (copies values from another object)

① Default Constructor :

A constructor with no parameters. It initializes data members with default values.

Syntax :

```
Class    class_name
  {
      public :
        // default constructor
      class_name ()
      {
          // constructor body
      }
  };
```

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// C++ program for default constructor

```
#include<iostream>
using namespace std;
class student
{
    public :

    int age;
    // default constructor
    student()
    {
        age = 20;
        cout<<"default constructor called"<<endl;
    }
};

int main()
{
    student s; // constructor is called automatically
    cout<<"age="<<s.age;
}
```

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② Parameterized Constructor:

A constructor that takes parameter to initialize the object with specific values.

Syntax:

```
class class_name
{
    public:
  // for one parameter

    class_name(data_type parameter1)
    {
        // Constructor body
    }
};
```

or

```
Class class_name
{
    public:
    // for more than one parameter
    class_name(data_type P1, data_type P2)
    {
        // Constructor body
    }
};
```

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// C++ program for one parameterized constructor

```
#include<iostream>
using namespace std;
class student
{
    public :
     int age;
    // parameterized conductor
      student ( int a)
     {
        age = a;
        cout<<"parameterized conductor called"<<endl;
     }
};

int main()
{
    student s(20); // constructor with parameter
    cout<<"age="<<s.age;
}
```

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// C++ program for more than one parameterized
// constructor

```
#include<iostream>
using namespace std;
class Student
{
    public:
     int age;
     int roll;
     string name;

    // parameterized Constructor

     Student (int a, int r, string n)
      {
          age = a;
          roll = r;
          name = n;
          cout<<" parameterized constructor called"<<endl;
      }
};
    int main()
    {
      // constructor with more than one parameterized
      Student s(20, 12355, "akash kumar");
      cout<<"age = "<<s.age<<endl;
      cout<<" roll no = "<<s.roll<<endl;
      cout<<" name="<<s.name;
    }
```

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// C++ program for more than one object [user ~~define~~ input]

```
#include<iostream>
using namespace std;
class student
{
    public:
        int age;
        int roll;
        string name;

    // parameterized constructor

    Student (int a, int r, int n)
    {
        age = a;
        roll = r;
        name = n;
        cout<<"parameterized constructor called"<<endl;
    }
};
int main()
{
    // taking used input data

    int age1, age2;
    int roll1, roll2;
    string name1, name2;
```

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```
cout<<"write age of 1st student";
cin>>age1;
cout<<"write roll of 1st student";
cin>>roll1;
cout<<"write name of 1st student";
cin>>name1;
cout<<"write age of 2nd student";
cin>>age2;
cout<<"write roll of 2nd student";
cin>>roll2;
cout<<"write name of 2nd student";
cin>>name2;

// Constructor with more than one parameter
// and more than one object

student s1(age1, roll1, name1); // 1st object
student s2(age2, roll2, name2); // 2nd object

cout<<"age of 1st = "<<s1.age;
cout<<"roll of 1st = "<<s1.roll;
cout<<"name of 1st = "<<s1.name;
cout<<"age of 2nd = "<<s2.age;
cout<<"roll of 2nd = "<<s2.roll;
cout<<"name of 2nd = "<<s2.name;
}
```

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```
// C++ program for parameterized constructor
// using member function

#include<iostream>
using namespace std;
class student
{
    public :
     int age;
     int roll;
     string name;

     // parameterized constructor

    Student (int a, int r, string n)
     {
        age = a;
        roll = r;
        name = n;
        cout<<"parameterized constructor called"<<endl;
     }


     void show() // using member function
      {
        cout<<"name="<<name<<endl;
        cout<<"age="<<age<<endl;
      }
};
```

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```
int main()
{
    // constructor with parameter
    student s(20, 12355, "akash kumar");
    s.show(); // function call
}
```

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